Fefferman Statement (B) on $\mathbb{T}^3$: A Precise Lean Formalization, a Kernel-Checked Reduction, and the Boundaries of the Frohmanian Symplectic Tether

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Abstract

This monograph records, in Lean 4 against Mathlib, a precise formal statement of Fefferman's Statement (B) on global smoothness of the unforced incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, together with an exact ledger of the analytic inputs that the Frohmanian symplectic tether does *not* supply. Statement (B) is *not* proved here. What is proved is a reduction: Statement (B) follows from the classical local theory with the Beale–Kato–Majda continuation criterion and from the open hypothesis that the Beale–Kato–Majda integral is finite on every finite interval of existence. We also record the two-dimensional clock test: constant vortex direction ($\sin\varphi \equiv 0$) coexists with linear growth $\|\phi(t)\|_{L^\infty} = M_0^2\, t$ on steady Euler equilibria.

1 Scope and non-claim boundary

This repository is a boundary and gap ledger. It is **not** a resolution of the Clay Millennium Problem. Every open input is a named proposition in Lean; none is assumed as an axiom, and no proof uses `sorry`. The axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. The active continuation of this programme, in which the remaining gap is decomposed into finer named statements, is `BenFrohman/periodic-ns-classical-continuation`.

- **Repository:** `https://github.com/BenFrohman/Clay-`
- **Concept DOI:** `10.5281/zenodo.23653012`
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2 Statement (B)

Following the official problem description of Fefferman [1], fix $\nu > 0$ and let u° be a smooth, divergence-free vector field on $\mathbb{R}^3$ with $u^\circ(x + e_j) = u^\circ(x)$ for $j = 1, 2, 3$. Take $f \equiv 0$.

> **Statement (B).** There exist smooth functions $p(x,t)$, $u_i(x,t)$ on $\mathbb{R}^3 \times [0,\infty)$ such that

$$\partial_t u_i + \sum_{j=1}^{3} u_j\, \partial_j u_i = \nu\, \Delta u_i - \partial_i p, \qquad \operatorname{div} u = 0, \qquad u(x,0) = u^\circ(x),$$

and u, p are 1-periodic in each spatial coordinate.

In Lean this is the proposition `Clay.StatementB.FeffermanStatementB` (file `lean/Clay/StatementB.lean`). Space is $\mathbb{R}^3$ realised as Euclidean space on Fin 3; partial derivatives are Fréchet derivatives along the standard basis; the time derivative is taken within $[0,\infty)$; smoothness is C^∞ on $[0,\infty)\times\mathbb{R}^3$. No component of the statement is opaque or trivial.

3 The proved reduction

Write $\omega = \operatorname{curl} u$. Consider the propositions

- `PeriodicMaximalSolutionWithBKM` (classical; Kato, Fujita–Kato, Beale–Kato–Majda [2]; not yet formalized): for all admissible data there are $T^* \in (0,\infty]$ and a smooth solution on $[0,T^*)$ such that $T^* < \infty$ implies $\int_0^{T^*} \|\omega(t)\|_{L^\infty}\,dt = \infty$;
- `PeriodicBKMIntegrability` (**open**; the Clay core): every smooth solution on a finite interval $[0,T)$ satisfies $\int_0^{T} \|\omega(t)\|_{L^\infty}\,dt < \infty$.

Theorem 1 (`feffermanStatementB_of_bkm`, kernel-checked)**.** *`PeriodicMaximalSolutionWithBKM` and `PeriodicBKMIntegrability` together imply Statement (B).*

Theorem 2 (`bkmIntegralFinite_of_holographic`, kernel-checked)**.** *If $\|\omega(t)\|_{L^\infty}$ is bounded on every finite interval of existence, then `PeriodicBKMIntegrability` holds.*

The second hypothesis of the first theorem is open and is equivalent in strength to Statement (B) itself. Consequently the reduction does not prove Statement (B).

4 Ledger of missing inputs

The file `lean/Clay/Gaps/MissingInputs.lean` states each former placeholder as a precise proposition:

- `metriplecticControlsEnstrophyL2`: $\sup_{[0,T)} \int_{\mathbb{T}^3} |\omega|^2 < \infty$ for every solution on a finite interval (open);
- `czGivesBMONotLinf`: no constant C gives $\|\nabla v\|_{L^\infty} \le C\|\operatorname{curl} v\|_{L^\infty}$ for smooth periodic divergence-free v (classical, not formalized);
- `spikeObstruction`: no constant C gives $\|f\|_{L^\infty} \le C\|f\|_{L^6(\mathbb{T}^3)}$ for smooth periodic f (classical, not formalized);
- `holographicNotAZFCBound`: the uniform ceiling $\sup_{[0,T)} \|\omega\|_{L^\infty} < \infty$ (open);
- `depletedStretching`: $\omega \cdot S\omega \le C|\omega|^2$ uniformly on finite intervals (open).

The former trivial definitions are retained under the suffix `_placeholder` and marked deprecated. Kernel-checked elementary results include the identity $\Pi_u u = 0$ for the pointwise tether projection, the unboundedness of the history integral $m^2 t$ for $m > 0$, and the conditional Riccati rearrangement.

References

[1] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, in: The Millennium Prize Problems, Clay Mathematics Institute, 2006, pp. 57–67.

[2] J. T. Beale, T. Kato, A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[3] P. Constantin, C. Fefferman, *Direction of vorticity and the problem of global regularity for the Navier–Stokes equations*, Indiana Univ. Math. J. **42** (1993), 775–789.

5 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/Clay-` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.js` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
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COMPARATOR.md	1198 B	a899cda8bd5c7160...
LICENSE	11355 B	9da72c2bee4f20ba...
NOTICE	317 B	80e5fa3638d62d2a...
README.md	903 B	d173785a4df45954...
STATUS.md	2491 B	252aae1e3a6a7243...
latex/geometric_boundaries.tex	5056 B	952c5705ca8851c5...
latex/section10_gaps.tex	2314 B	c43c9ac50665ba92...
latex/two_d_fact.tex	1490 B	b969949b2b80c90d...
lean/Clay/Depletion/GeometricAlignment.lean	490 B	4d7125ee4bc18711...
lean/Clay/Gaps/BKMIntegral.lean	511 B	19cea215acbc7903...
lean/Clay/Gaps/ConditionalRiccati.lean	693 B	b3c47df1a1cbeb43...
lean/Clay/Gaps/DirectionAlignment.lean	633 B	3395eae5672c0273...
lean/Clay/Gaps/HistoryClock.lean	662 B	b134ea98037b29fa...
lean/Clay/Gaps/MissingInputs.lean	795 B	010aed7022d8fcd7...
lean/Clay/Gaps/RejectedMeasureBound.lean	531 B	da60d8abc07fa028...
lean/Clay/Gaps/StretchingIntegral.lean	765 B	a1b919459c9bf774...
lean/Clay/Gaps/TwoDClock.lean	935 B	09bf6a419006c185...
lean/Clay/Gaps.lean	422 B	99171b435e416ff6...
lean/Clay/Tether.lean	1031 B	2e562a881a4d654a...
lean/Clay.lean	477 B	1d60fd986c2b3aba...
lean/lakefile.lean	577 B	3d3b19f370345872...
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